

Designing for Fair Oracle Extractable Value: A Theoretical Framework and Empirical Findings from DeFi

Nicolas Andreoulis¹, Marco Di Maggio², Louis-Henri Merino³, Kyle Montag¹,
and James Ward¹

¹ Chainlink Labs

² Imperial College Business School

³ École Polytechnique Fédérale de Lausanne

Abstract. This paper studies Oracle Extractable Value (OEV) as a distinct channel of rent and risk in decentralized finance (DeFi). Using granular data on liquidations from Aave V2 and V3 across Ethereum and major rollups (2023–2025), we document that OEV is large, heavy-tailed, and increasingly captured by vertically integrated builder–searcher coalitions, especially during periods of market stress. We then introduce a stylized model of Oracle Update Rebate Window (OURW), an oracle mechanism that seeks to redirect a portion of liquidation profits to protocols. The model establishes conditions under which censorship is unprofitable and shows how refund rates, fallback windows, and builder competition interact to deter attacks. A case study of Aave illustrates how these dynamics manifest empirically. Together, our analysis integrates descriptive evidence and mechanism design to provide practical guidance: with moderate parameters and sufficient builder diversity, OURW-style mechanisms can safely recover value while preserving liveness.

1 Introduction

Decentralized finance (DeFi) lending platforms such as Aave [1] and Compound [9] allow users to borrow against crypto collateral without centralized intermediaries. When the value of posted collateral falls below a safety threshold, the protocol triggers an automatic liquidation. This process depends crucially on price updates delivered by external oracles ([23], [24]). Once oracle prices move positions into debt, any participant can repay the debt and seize collateral at a discount. Because the rewards accrue to the highest paying actor within a block time-window, liquidations resemble ultra-competitive races, where automated bots compete to extract profit margins [17]. These profits fall under the broader category of Maximal Extractable Value (MEV) (see [11] and [21]), defined as the value derived from strategically ordering, including, or censoring transactions within a block.

The analogy to traditional markets is illustrative. Imagine a distressed debt auction triggered automatically when a third-party data provider marks down asset quality. Whoever reacts first purchases the collateral at below-market prices.

Now extend this analogy: some participants not only compete in the auction but also control the auctioneer itself. In DeFi, such vertical integration occurs when the same entity both constructs blocks and operates liquidation bots. This dual role transforms a competitive market into a censorship problem, since block builders can exclude rival transactions and strategically time inclusion to maximize profit ([20] and [26]).

The rising magnitude of liquidation-driven MEV has prompted protocols to explore whether a share of this value can be redirected to users or treasuries [10, 15]. We study a class of mechanisms we call Oracle Update Rebate Window (OURW), of which Smart Value Recapture (SVR) [6] is one prominent example. Under OURW, oracle price updates are routed through a private channel for a short window. If a liquidation occurs during this interval, the liquidator must refund a fraction of profits back to the protocol. If no liquidation occurs, or the private update fails to land on-chain, the system falls back to a public feed with no fee. The fallback ensures liveness, but it also creates a strategic incentive for a vertically integrated builder-liquidator to withhold the private update, wait for the fallback, and thereby avoid the refund. This design, whilst seemingly net beneficial for protocols, doesn't guarantee a reduction in rent from MEV participants. A vertically integrated builder-liquidator may censor the private update, wait for the fallback, and thereby avoid the refund.⁴ The viability of this censorship hinges on the value of the avoided fee, the opportunity cost of sustaining censorship, and the additional gains from bundling multiple liquidations in a single block.

This paper is the first to provide an integrated theoretical and empirical framework for understanding censorship threats to OURW-style mechanisms. We combine detailed on-chain data with a set of formal models to analyze when censorship is profitable and how design choices can deter it. Our contributions unfold in two layers. First, we establish empirical regularities about Oracle Extractable Value (OEV), the subcategory of MEV originating specifically from oracle-driven state changes such as liquidations. Second, we develop mechanism-design models showing how censorship incentives depend on refund rates, fallback windows, and builder competition.

Empirically, our dataset covers liquidations from Aave V2 and V3 across Ethereum and major rollups between 2023 and mid-2025. We document five stylized facts. OEV is large and extremely concentrated, with a handful of crisis periods accounting for the majority of profits. The share of profits accruing to builders has risen sharply, surpassing 60% by 2025 and over 90% during stress events, highlighting the importance of vertical integration. Builder competition acts as a natural deterrent: as competition rises, censorship rents dissipate. OEV opportunities correlate strongly with downside price shocks and blockspace congestion, rendering attack surfaces predictable. Finally, fallback markets are often

⁴ It is worth noting that builder-liquidators still have the ability to profit more than non vertically integrated searchers, as they keep the 10% builder split built into the auction. This also allows them to outbid other competitors to ensure they capture revenue.

thin, amplifying the feasibility of censorship when searcher concentration is high. These observations ground our subsequent modeling choices.

Our theoretical analysis begins with a static model of a single liquidation opportunity. The protocol chooses a refund rate and fallback window. We derive a threshold condition: if the refund rate remains below this boundary, censorship is strictly unprofitable. This simple mechanism-design insight shows that censorship resistance can be engineered *ex ante* by calibrating two tunable parameters. Extending beyond this baseline, we model censorship as a multi-block contest among builders. Using a Tullock-style all-pay contest formulation [8], we show that even modest builder diversity significantly raises the cost of censorship, often beyond the gains from avoiding the refund. The implication is that market structure—specifically builder concentration—is as important as protocol parameters in determining censorship resistance.

We then consider dynamic environments where liquidation sizes follow Pareto-like heavy tails. Here, censorship is not continuous but episodic, targeting only the largest events. The model predicts selective exclusion strategies: attackers rationally censor a handful of extreme liquidations, not routine ones. Protocol design levers such as longer fallback windows, higher fixed costs of censorship, or faster expectation updates reduce incentives for such tail censorship. The framework therefore explains why censorship is observed only during crises and highlights how protocols can preemptively adapt.

This work connects to several strands of literature. The first is blockchain MEV. Early studies by Daian et al. [11] documented frontrunning and arbitrage in Ethereum’s decentralized exchanges. Subsequent work expanded MEV taxonomies across financial primitives [2, 18]. Yet liquidation-based MEV triggered by oracle updates has received less attention, and censorship incentives in private oracle channels remain underexplored. By analyzing OURW-specific vulnerabilities, our study fills this gap.

A second strand is mechanism design under limited commitment and exclusion. Classical results [13] emphasize how weak enforcement shapes incentives. See the survey by [3] for an overview of this literature. In the OURW case, if builders exclude oracle updates long enough, the refund mechanism collapses. Our static model provides conditions under which honest behavior dominates, offering a tractable design principle for censorship resistance.

The third strand is contest theory and dynamic rent-seeking. Tullock [19] pioneered models of probabilistic contests where resource allocation is determined by relative effort. Later extensions [16], [14] emphasized dissipation under heterogeneity. In our setting, builder competition operates as a contest for block inclusion, with bribes and opportunity costs endogenized. As builder diversity increases, rents dissipate, deterring censorship. Our dynamic extension further connects to repeated rent-seeking models ([12], [22]), where persistence and heavy-tailed shocks govern equilibrium. We show that selective censorship of extreme liquidations aligns closely with these theoretical predictions.

Finally, our analysis speaks to the political economy of consensus-layer economics. Prior work by Budish, Cramton, and Shim [4] argued that latency

racers in financial markets are a design failure, not a technological inevitability. Blockchain researchers have proposed encrypted mempools [7, 25] and sealed-bid block auctions ([5]) to mitigate similar inefficiencies.

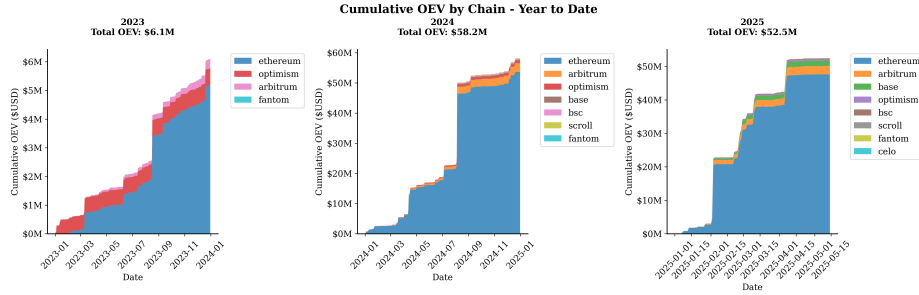
In sum, this paper provides both empirical insight and theoretical tools to evaluate OEV-related censorship threats. It contributes to MEV research by highlighting a new vulnerability channel, to mechanism design by formalizing conditions for incentive-compatible OURW, and to contest theory by embedding censorship in a dynamic rent-seeking framework. By integrating on-chain evidence with mechanism design, we aim to offer actionable guidance for protocols seeking to capture value while preserving system integrity. The remainder of the paper presents empirical descriptives of OEV (Section 2), develops a toy model of censorship (Section 3), and applies the framework to Aave as a case study (Section 4), before concluding with broader design and policy implications.

2 Empirical Facts and Descriptives about OEV

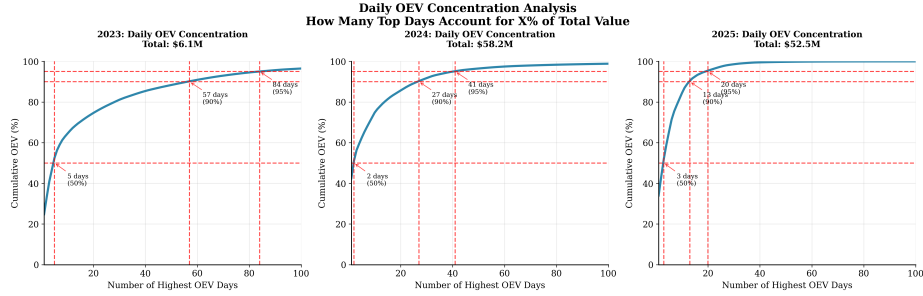
We define *Oracle Extractable Value* (OEV) as the component of Maximal Extractable Value that arises specifically from oracle-mediated state changes. In lending protocols, these events are almost always liquidations, triggered by an adverse oracle price update. OEV thus differs from other MEV channels such as arbitrage or sandwiching, because its existence is tightly linked to oracle design and its timing is predictable at the moment of an update. Our empirical analysis draws on a comprehensive dataset of liquidations across Aave V2 and V3 from January 2023 through May 2025, covering Ethereum as well as major rollups including Arbitrum, Optimism, and Base. For each event we compute the liquidation surplus attributable to the oracle update, identify the winning searcher and builder, and record contemporaneous gas conditions and asset pairs. This allows us to construct a transaction-level view of the size, distribution, and market structure of OEV.

2.1 Scale and Concentration

Although there was significant OEV potential in the aftermath of the DeFi summer, total OEV grew from approximately \$6 million in 2023 to over \$50 million by mid-2025, with Ethereum consistently accounting for more than ninety percent of value. Figure 1 illustrates that Ethereum dominates both the cumulative and relative shares of OEV, while rollups contribute only a small fraction. The distribution of liquidation profits is extremely skewed. A handful of events account for the majority of value: fewer than one percent of liquidations produce over eighty percent of aggregate OEV. In 2024, two trading days alone generated more than half of all liquidation profits, underscoring the heavy-tailed nature of the distribution. Table 1 shows that the 99.9th percentile liquidation exceeds five orders of magnitude above the median. This concentration motivates our later theoretical focus on tail censorship.



(a) Year-to-date cumulative OEV stacked by chain. Ethereum drives virtually the entire series: its step-ups coincide with the crash windows highlighted earlier, while roll-ups contribute only thin “icing” layers. The widening gap over time underscores why our builder-competition results generalise primarily to Ethereum.



(b) Cumulative share of annual OEV captured by the highest-value days. In 2024 only *two* days account for 50% of liquidation value and 27 days cover 90%—quantitative evidence that any censorship attempt would rationally target a tiny set of crisis blocks.

Fig. 1: OEV event timing around oracle price updates. Ethereum dominates across years, with rollups contributing a thin margin.

2.2 Builder Ascendancy

A key empirical trend is the redistribution of profits from searchers to builders. In early 2023, most liquidation value accrued to searchers submitting transactions. By mid-2025, builders captured between sixty and seventy percent of profits on average, and more than ninety percent during extreme stress episodes. This transition produces a U-shaped distribution of profit shares across liquidation sizes. Small and mid-sized events remain relatively competitive, but both the smallest and the largest liquidations are dominated by builders. The former reflects minimum fixed costs of block inclusion, while the latter reflects vertical integration and coordination during extreme stress. Figures 2 and 6 confirm that the builder share rises steadily over time and accelerates precisely in the tail.

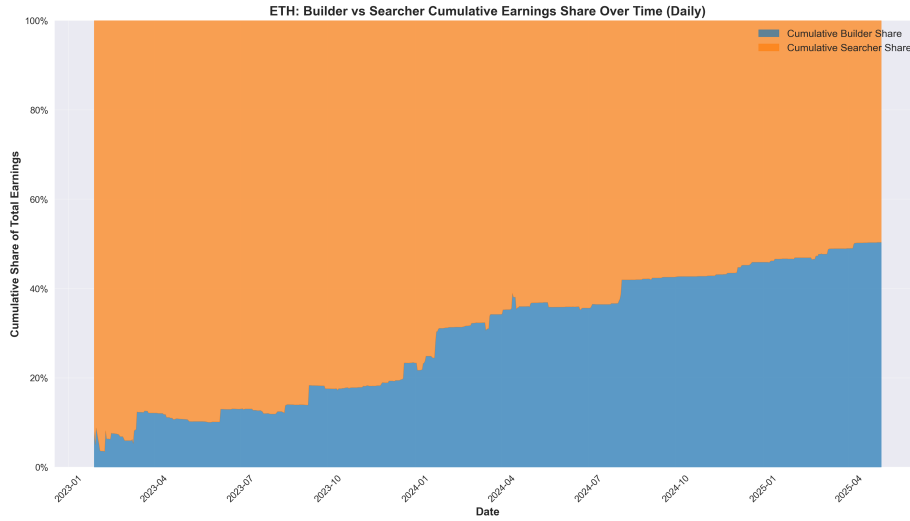


Fig. 2: Builder share of cumulative earnings on Ethereum rises from $\approx 5\%$ in early-2023 to nearly 50% by mid-2025. The smooth upward trajectory reinforces the interpretation that builders retain more value the longer the PBS market matures and concentrates.

Fig. 3: Distributional overview of OEV-related opportunities. Builder capture rises sharply in the tails.

2.3 Role of Builder Competition

We proxy competition using the inverse Herfindahl–Hirschman Index of daily builder market shares. As competition increases, builder profits decline and searcher profits rise, consistent with rent dissipation. Time-series regressions confirm that a one standard deviation increase in competition reduces builder profit share by several percentage points the following day. This validates the theoretical intuition that censorship becomes prohibitively costly in competitive builder environments. Conversely, concentration amplifies the feasibility of censorship, particularly when one or two builders dominate block production.

2.4 Predictable Attack Surface

Censorship opportunities are not uniformly distributed but correlate with observable conditions. Downside price shocks explain far more of the variation in OEV than do volatility measures alone. When collateral assets experience sharp drawdowns, clustered liquidations arise, creating the potential for bundled profits. Simultaneously, congestion in the blockspace market, proxied by the base fee per gas, is strongly correlated with high-OEV events. These conditions are

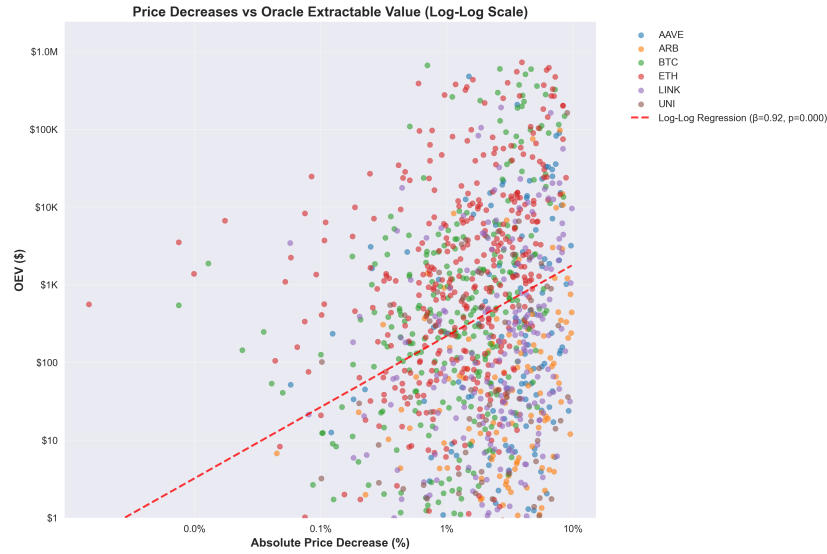


Fig. 4: Absolute intraday price *decreases* (%) against real-time oracle-extractable value (OEV), log-log scale. The dashed red line fits a log-log regression whose slope is ($\beta \approx 0.92, p < 0.001$). Liquidation windfalls therefore scale nearly proportionally with downside moves. Yet significant OEV still arises on sub-1 % drawdowns, underscoring that oracle-update latency remains exploitable even during relatively muted sell-offs.

predictable in real time, suggesting that sophisticated integrated actors can anticipate when censorship would be most rewarding. Figure 4 depicts how OEV spikes align with episodes of price collapse and gas surges.

2.5 Thin Fallback Markets

Finally, our analysis shows that fallback liquidation markets are thin and concentrated. In many episodes, a small number of searchers dominate participation. When one searcher exceeds fifty-five percent of profit share, builders capture nearly all residual rents, either through tacit coordination or outright integration. This undermines the assumption that fallback liquidations are competitive, increasing the attractiveness of censorship. In practice, this means that when OURW windows expire without action, the liquidation outcome is likely to be monopolized rather than contested.

2.6 Summary of Stylized Facts

To summarize, five facts stand out. First, OEV is large, growing, and highly concentrated in a few extreme events. Second, builders increasingly capture the

Table 1: Summary of OEV-adjacent events on Aave. Extreme concentration across a handful of events.

surplus, especially in tails. Third, builder competition dissipates censorship rents. Fourth, high-OEV opportunities are predictable, coinciding with price crashes and gas congestion. Fifth, fallback markets are thin and dominated by a few actors, amplifying risk. Together, these findings motivate our subsequent models. They suggest that censorship incentives are episodic rather than continuous, that market structure is central to feasibility, and that protocol design must take both into account. These stylized facts are consistent with the intuition that censorship is most threatening during crisis periods, precisely when liquidation liveness is most critical.

In the next section, we formalize these insights in a toy model of censorship, showing how refund rates and fallback windows interact with builder concentration to shape equilibrium incentives. This model provides the theoretical backbone for interpreting the empirical patterns documented here.

3 A Simple Toy Model of Censorship

To formalize the strategic incentives surrounding Private-Update Value Recapture (OURW), we begin with a stylized model of a single liquidation opportunity. The protocol implements OURW with two tunable parameters: the refund rate $r \in [0, 1]$, which specifies what fraction of the liquidation bonus must be refunded to the protocol if the private channel is used, and the fallback window $X \in \mathbb{N}$, which determines the number of blocks before the system defaults to a public feed with no fee. An integrated attacker, combining a liquidation bot with block-building power, observes (r, X, L) and decides whether to (1) liquidate honestly within the fallback window and pay r , or (2) censor inclusion of the private oracle update until the fallback activates and then liquidate with no refund.

3.1 Payoff Structure

Let $L > 0$ denote the liquidation value and $\alpha \geq 1$ a synergy factor capturing the added benefit from bundling multiple liquidations. An integrated attacker controls a fraction θ of block production. Honest execution yields a payoff of $(1-r)\alpha L$, net of any competitive bid. By contrast, censorship allows the attacker to retain the full αL but requires sustained exclusion for X blocks. The cost of censorship grows with $(1-\theta)$, the share of blocks controlled by others, and with the length of the fallback window. Formally, censorship is profitable if

$$\alpha L - C(X) > (1-r)\alpha L, \quad (1)$$

where $C(X)$ is the cumulative cost of excluding the OURW message.

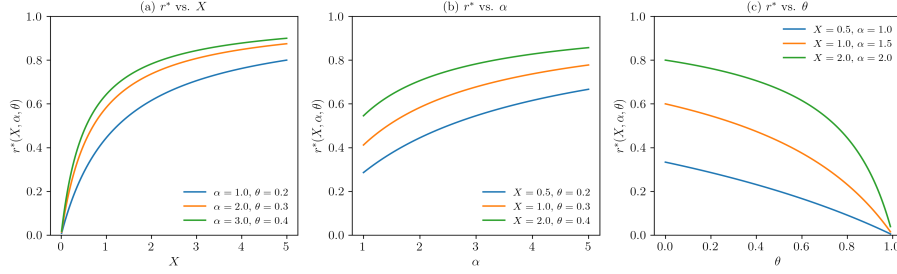


Fig. 5: Toy model comparative statics (qualitative schematic). The censorship thresholds r^* and $r^\dagger$ rise with fallback window X and synergy α , but fall with attacker block power θ and decline further when builder competition increases.

Assuming bribes to honest builders equal their foregone OURW revenue, $C(X)$ scales linearly in X , $(1 - \theta)$, and $(1 - r)\alpha L$. The resulting static threshold condition is

$$r^* = \frac{X(1 - \theta)\alpha}{1 + X(1 - \theta)\alpha}. \quad (2)$$

If the protocol sets $r < r^*$, honest liquidation dominates; if $r > r^*$, censorship becomes attractive.

3.2 Comparative Statics

This condition yields intuitive insights, as depicted in fig. 5. A longer fallback window raises r^* , making censorship harder. Greater liquidation synergies α increase r^* , permitting more aggressive recapture without risk. Higher block share θ lowers r^* , reflecting the heightened vulnerability under concentrated block production. These comparative statics align with empirical evidence: censorship is most plausible during crash periods when liquidations are bundled and builder concentration is high.

3.3 Extensions: Builder Competition

The static model treats bribe costs as exogenous. In practice, builders compete in an all-pay contest for block inclusion. Embedding censorship in a Tullock-style framework yields a contest-based profitability boundary:

$$r^\dagger = \frac{X(n - 1)}{n + X(n - 1)}, \quad (3)$$

where $n \geq 2$ is the number of competing builders. This contest threshold $r^\dagger$ is (weakly) lower than the static threshold r^* whenever $n > 1$, shrinking the region in which censorship is profitable. As $n \rightarrow \infty$, we have $r^\dagger \rightarrow 1$, i.e., builder diversity fully dissipates censorship rents and makes censorship strictly dominated.

3.4 Interpretation

The toy model underscores two central points. First, censorship resistance can be engineered ex ante through simple parameter choices. Second, censorship incentives are episodic rather than continuous, depending on liquidation size and market structure. The introduction of the contest threshold $r^\dagger$ highlights the protective role of builder competition: even modest diversity in block production sharply raises the expected cost of censorship, aligning the theory with the empirical evidence that integrated actors dominate only in tail events.

4 Aave Case Study

To illustrate how censorship incentives manifest in practice, we examine Aave, the leading DeFi lending protocol by liquidation volume. Our analysis uses transaction-level data from Aave V2 and V3 spanning January 2023 to May 2025, focusing on Ethereum where more than ninety percent of observed Oracle Extractable Value (OEV) arises. We track liquidation size, asset pairs, builder and searcher identities, and contemporaneous gas conditions. This enables us to connect the theoretical framework of the previous section to concrete market outcomes.

4.1 Empirical Patterns

Our dataset confirms that Aave liquidations are heavily skewed. Fewer than one percent of events account for more than eighty percent of total OEV. The largest liquidations, often exceeding hundreds of thousands of dollars in profit, cluster during sharp market downturns. Builders capture a growing share of these windfalls: by mid-2025 they routinely account for more than two-thirds of liquidation value, and over ninety percent in crash periods. This validates our model’s focus on integrated actors as the primary censorship threat.

Builder competition strongly shapes these outcomes. During periods with high competition, measured by inverse Herfindahl–Hirschman Index, builder rents fall and searchers capture greater value (fig. 6). When competition thins and one or two builders dominate, censorship becomes more feasible. Episodes of concentrated fallback participation exacerbate this dynamic: when a single searcher accounts for more than half of profits, builders appropriate nearly all residual value.

We also estimate the price changes required during oracle delays to eliminate profits altogether. These values represent the net impact on prices, accounting for both the delay and the change in liquidity. Price changes directly influence the potential bad debt for each scenario.

The profit per asset is given by:

$$\text{Profit per Asset} = \text{Total OEV}^{\text{adj}} - \text{Total Bad Debt per asset}_d^j.$$

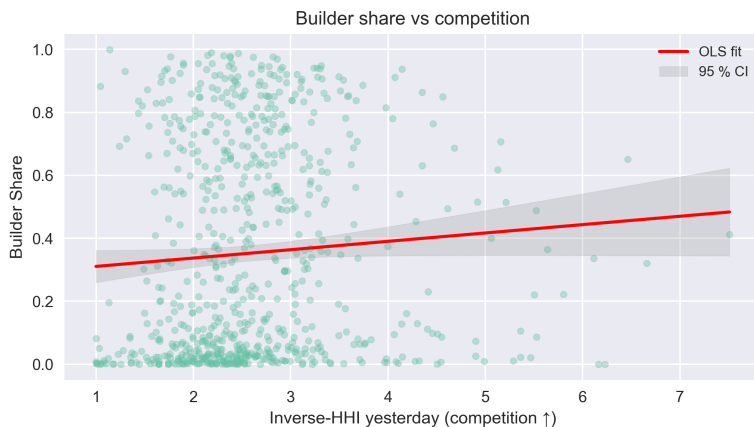


Fig. 6: The corresponding OLS for builders shows a positive and statistically significant slope: each unit increase in inverse-HHI lifts builder share by roughly four percentage points on average, in line with the theoretical elasticity $\partial r^\dagger / \partial n > 0$

To eliminate profits (Profit per Asset = 0), the required price change can be inferred from the historical data as:

$$\text{Price Change}_d^j = \text{Bad Debt}_d^{-1}(j), \quad j \in \{+, -\}.$$

As shown in Figure 7, eliminating OEV profits would require sizeable price movements. On the downside, WBTC would have to fall by about 9.8%, ETH by 7.8%, AAVE by 11.2%, and LINK by an extreme 79%. On the upside, WBTC would need to rise by 7.4%, ETH by 43.8%, and LINK by 79.5%. These thresholds indicate the magnitude of price changes necessary to erase OEV profits for each asset under both scenarios, highlighting that the existing liquidation bonuses provide a substantial buffer even under severe market conditions.

4.2 Qualitative Findings

Three qualitative insights emerge. First, censorship risk is selective. Even among Ethereum liquidations, attempts would rationally target only the very largest opportunities. Routine liquidations are too small to justify sustained exclusion over multiple blocks. This selectivity explains why censorship is episodic and crisis-driven rather than continuous.

Second, protocol parameters matter. Longer fallback windows increase the cost of sustained censorship and align with our theoretical threshold condition. Empirically, windows of three to six blocks appear sufficient to deter attacks under observed competition levels. Conversely, overly aggressive refund rates risk incentivizing censorship, particularly when builder concentration is high.

Third, market structure is decisive. Builder integration with searchers correlates strongly with censorship feasibility. In periods when builders remain frag-

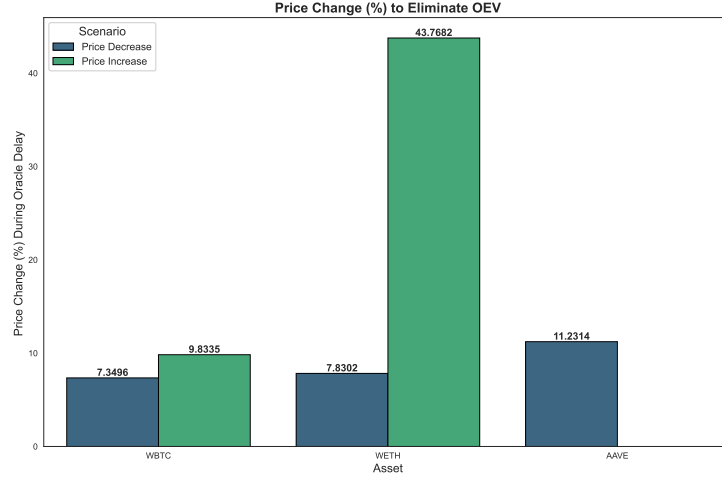


Fig. 7: Price change required during the oracle-update delay to drive liquidation profits (OEV) to zero, by asset. Blue bars denote the needed downside move; green bars denote the needed upside move. Eliminating OEV typically requires sizable moves—about 7–11% on the downside for WBTC/WETH/AAVE and much larger upside shocks ($\approx 10\%$ for WBTC, $\approx 44\%$ for WETH)

mented, censorship rents dissipate through bidding wars. When builders are vertically integrated or few in number, censorship incentives spike.

4.3 Implications for Governance

For Aave governance, these findings suggest that censorship resistance is attainable with careful calibration. Moderate refund rates, fallback windows of several blocks, and active promotion of builder competition substantially mitigate attack incentives. Importantly, the results highlight that censorship risk is not a pervasive daily hazard but a tail phenomenon coinciding with crisis events. Protocol design should therefore prioritize robustness during stress rather than maximizing revenue in normal times.

4.4 Interpretation

The Aave case study provides an empirical anchor for the toy model. It demonstrates that the theoretical threshold conditions map onto observed market behavior, and that censorship risk aligns with extreme liquidation events and builder concentration. While our redacted analysis withholds exact numerical calibrations and detection heuristics, the qualitative patterns are clear: censorship is possible but not pervasive, and its feasibility depends on protocol design choices and market structure. These insights reinforce the broader message that

OEV can be partially recaptured without compromising liveness, provided mechanisms account for the episodic and selective nature of censorship.

5 Discussion and Concluding Remarks

The empirical regularities and the toy models form a coherent picture of Oracle Extractable Value (OEV) dynamics in lending liquidations. First, OEV is episodic and heavy-tailed: a small number of crisis windows accounts for the majority of value. This aligns with the static threshold logic, where the benefit of censorship scales with liquidation size while per-block costs are relatively stable. Second, market structure is a first-order determinant of feasibility. The contest formulation shows that even modest builder diversity sharply raises the expected cost of sustained exclusion, tightening the safe region for the recapture rate.

A practical implication is that censorship risk concentrates exactly when protocols most need liveness: during sharp drawdowns and congestion. Configurations that are safe in average conditions may become brittle in the tail unless they incorporate margin—via longer fallback windows, moderate refund rates, or higher fixed costs for censorship. A complementary lever is to reduce entry frictions for non-integrated searchers, which thins the attack’s ex-post payoff and improves fallback competitiveness.

Limitations of our approach include the abstraction to a single-asset, single-chain baseline; the reduced-form treatment of some frictions (e.g., latency, relay quirks, alternative MEV); and deliberate redaction of implementation details. Nevertheless, the qualitative comparative statics are robust to these modeling choices, and the Aave evidence corroborates the core mechanisms at play.

Overall, our analysis shows that censorship resistance can be engineered ex ante with a handful of levers. Refund rates should remain at or below the safe region implied by the static threshold r^* or the stricter contest bound $r^\dagger$ when builder concentration is material. Fallback windows should be lengthened or randomized to raise the multi-block cost of exclusion. Builder diversity and searcher entry should be actively promoted so that rents dissipate in contests rather than accrue to censoring coalitions. When borrower welfare or systemic liveness is paramount, protocols should prioritize robustness in tail events rather than maximizing expected recapture in calm markets.

These principles are intentionally implementation-agnostic. They apply across oracle providers and chains wherever liquidation logic depends on timely price propagation. Future work should endogenize builder market formation, analyze cross-chain liquidation timing, and explore governance of dynamic fee schedules that adapt to market structure while resisting manipulation.

References

1. Aave: Aave Protocol Documentation, <https://docs-aave.vercel.app/docs/developers/liquidations>

2. Angeris, G., Chitra, T., Diamandis, T., Kulkarni, K.: The Specter (and Spectra) of Miner Extractable Value (Oct 2023). <https://doi.org/10.48550/arXiv.2310.07865>
3. Bergemann, D., Välimäki, J.: Dynamic mechanism design: An introduction. *Journal of Economic Literature* **55**(2), 431–492 (2017). <https://doi.org/10.1257/jel.20151343>
4. Budish, E., Cramton, P., Shim, J.: The High-Frequency Trading Arms Race: Frequent Batch Auctions as a Market Design Response. *The Quarterly Journal of Economics* **130**(4), 1547–1621 (Nov 2015), <https://doi.org/10.1093/qje/qjv027>
5. Capponi, A., Cartea, Á., Drissi, F.: Do longer block times impair market efficiency in decentralized markets? *SSRN Electronic Journal* (Jun 2025). <https://doi.org/10.2139/ssrn.5290232>, <https://ssrn.com/abstract=5290232>
6. Chainlink: Introducing SVR: A Chainlink-Powered MEV Recapture Solution For DeFi (Dec 2024), <https://blog.chain.link/chainlink-smart-value-recapture-svr/>
7. Choudhuri, A.R., Garg, S., Piet, J., Policharla, G.V.: Mempool Privacy via Batched Threshold Encryption: Attacks and Defenses. pp. 3513–3529 (2024), <https://www.usenix.org/conference/usenixsecurity24/presentation/choudhuri>
8. Chowdhury, S.M., Sheremeta, R.M.: A generalized Tullock contest. *Public Choice* **147**(3), 413–420 (Jun 2011), <https://doi.org/10.1007/s11127-010-9636-3>
9. Compound: Compound III Docs | Liquidation, <https://docs.compound.finance/liquidation/>
10. Compound DAO: Request for Proposal (RFP): Oracle Extractable Value (OEV) Solution for Compound Protocol - Protocol Development (May 2025), <https://www.comp.xyz/t/request-for-proposal-rfp-oracle-extractable-value-oev-solution-for-compound-protocol/6786>
11. Daian, P., Goldfeder, S., Kell, T., Li, Y., Zhao, X., Bentov, I., Breidenbach, L., Juels, A.: Flash Boys 2.0: Frontrunning in Decentralized Exchanges, Miner Extractable Value, and Consensus Instability. In: 2020 IEEE Symposium on Security and Privacy (SP). pp. 910–927 (May 2020), <https://ieeexplore.ieee.org/document/9152675>
12. Gradstein, M., Konrad, K.A.: Orchestrating Rent Seeking Contests. *The Economic Journal* **109**(458), 536–545 (1999). <https://doi.org/10.1111/1468-0297.00461>
13. Green, J., Laffont, J.J.: On Coalition Incentive Compatibility. *The Review of Economic Studies* **46**(2), 243–254 (1979). <https://doi.org/10.2307/2297048>
14. Hillman, A.L., Riley, J.G.: Politically Contestable Rents and Transfers. *Economics & Politics* **1**(1), 17–39 (1989). <https://doi.org/10.1111/j.1468-0343.1989.tb00003.x>
15. Moonwell DAO: Evaluating OEV Solutions for Moonwell: Community Temperature Check - Proposals / Moonwell Improvement Proposal (Dec 2024), <https://forum.moonwell.fi/t/evaluating-oev-solutions-for-moonwell-community-temperature-check/1456>
16. Nitzan, S.: Modelling rent-seeking contests. *European Journal of Political Economy* **10**(1), 41–60 (1994), <https://EconPapers.repec.org/RePEc:eee:poleco:v:10:y:1994:i:1:p:41-60>
17. Qin, K., Zhou, L., Gamito, P., Jovanovic, P., Gervais, A.: An empirical study of DeFi liquidations: incentives, risks, and instabilities. In: Proceedings of the 21st ACM Internet Measurement Conference. pp. 336–350. IMC '21, Association for Computing Machinery, New York, NY, USA (Nov 2021). <https://doi.org/10.1145/3487552.3487811>
18. Qin, K., Zhou, L., Gervais, A.: Quantifying Blockchain Extractable Value: How dark is the forest? pp. 198–214. IEEE Computer Society (May 2022). <https://doi.org/10.1109/SP46214.2022.9833734>

19. Tullock, G.: The Welfare Costs of Tariffs, Monopolies, and Theft. *Economic Inquiry* **5**(3), 224–232 (1967). <https://doi.org/10.1111/j.1465-7295.1967.tb01923.x>
20. Wahrstätter, A., Ernstberger, J., Yaish, A., Zhou, L., Qin, K., Tsuchiya, T., Steinhorst, S., Svetinovic, D., Christin, N., Barcentewicz, M., Gervais, A.: Blockchain Censorship. In: *Proceedings of the ACM Web Conference 2024*. pp. 1632–1643. WWW '24, Association for Computing Machinery, New York, NY, USA (May 2024). <https://doi.org/10.1145/3589334.3645431>
21. Yang, S., Zhang, F., Huang, K., Chen, X., Yang, Y., Zhu, F.: SoK: MEV Countermeasures. In: *Proceedings of the Workshop on Decentralized Finance and Security*. pp. 21–30. DeFi '24, Association for Computing Machinery, New York, NY, USA (Nov 2024). <https://doi.org/10.1145/3689931.3694911>
22. Yildirim, H.: Contests with multiple rounds. *Games and Economic Behavior* **51**(1), 213–227 (Apr 2005). <https://doi.org/10.1016/j.geb.2004.03.004>
23. Zhang, F., Cecchetti, E., Croman, K., Juels, A., Shi, E.: Town Crier: An Authenticated Data Feed for Smart Contracts. In: *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*. pp. 270–282. CCS '16, Association for Computing Machinery, New York, NY, USA (Oct 2016). <https://doi.org/10.1145/2976749.2978326>
24. Zhang, F., Maram, D., Malvai, H., Goldfeder, S., Juels, A.: DECO: Liberating Web Data Using Decentralized Oracles for TLS. In: *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security*. pp. 1919–1938. CCS '20, Association for Computing Machinery, New York, NY, USA (Nov 2020). <https://doi.org/10.1145/3372297.3417239>
25. Zhang, H., Merino, L.H., Qu, Z., Bastankhah, M., Estrada-Galiñanes, V., Ford, B.: F3B: A Low-Overhead Blockchain Architecture with Per-Transaction Front-Running Protection. In: Bonneau, J., Weinberg, S.M. (eds.) *AFT 2023*. vol. 282, pp. 3:1–3:23 (2023). <https://doi.org/10.4230/LIPICS.AFT.2023.3>
26. Öz, B., Sui, D., Thiery, T., Matthes, F.: Who Wins Ethereum Block Building Auctions and Why? In: Böhme, R., Kiffer, L. (eds.) *6th Conference on Advances in Financial Technologies (AFT 2024)*. *Leibniz International Proceedings in Informatics (LIPIcs)*, vol. 316, pp. 22:1–22:25. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl, Germany (2024). <https://doi.org/10.4230/LIPICS.AFT.2024.22>